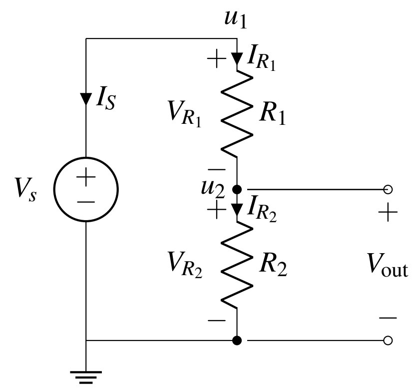
EECS 16A Designing Information Devices and Systems I
Discussion 7A

1. Voltage Divider

For the circuit below, your goal will be to find the voltage V_{out} in terms of the resistances R_1, R_2, and V_s, using NVA (Node Voltage Analysis). The labeling steps (steps 1-4) have already been done for you.

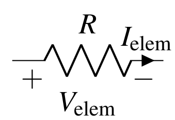


Here is a reminder of the labeling steps followed to get the circuit diagram above:

- **Step 1:** Select a reference (ground) node. Any node can be chosen for this purpose. We will measure all of the voltages in the rest of the circuit relative to this point.
- **Step 2:** Label all nodes with voltage set by voltage sources.
- **Step 3:** Label remaining nodes.
- **Step 4:** Label element voltages and currents, following **Passive Sign Convention** (discussed below).

Passive sign convention

The **passive sign convention** dictates that positive current should *enter* the positive terminal and *exit* the negative terminal of an element. Below is an example for a resistor:



As long as this convention is followed consistently, it does not matter which direction you arbitrarily assigned each element current to; the voltage referencing will work out to determine the correct final sign. When we discuss *power* later in the module, you will see why we call this convention "passive."

To achieve your goal of ***finding*** V_{out}, perform the rest of the NVA steps in the boxes below:

Step 5: Write KCL equations for all nodes with unknown voltages.

Step 6: Find expressions for all element currents in terms of element voltages and characteristics.

Step 7: Substitute all element voltages in your step 6 equations with node voltages.

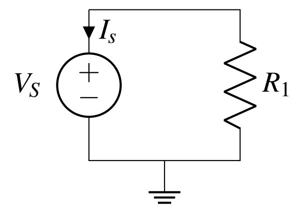
Step 8: Substitute expressions found in step 7 into the KCL equations from step 5.

Step 9: Solve for the node voltage values. At this point the analysis procedure is effectively complete - all that's left to do is solve the system of linear equations (by applying Gaussian Elimination, inverting $\mathbf{A}$, direct substitution, etc.) to find the values for the u's. Then we can go back to our Step 7 equations and calculate the I's. Note that in our circuit, $V_{out} = u_2 - 0 = u_2$.

2. A Simple Circuit

Use KVL and/or KCL to solve the following circuits.

(a) For this problem assume $V_S = 1V$ and $R_1 = 1k\Omega$. Find the current, I_s flowing through the voltage source.



(b) For this problem assume $V_S = 1V$, $R_1 = 2k\Omega$, and $R_2 = 2k\Omega$. Find the current, I_s flowing through the voltage source.